**SCIENCE PROGRAM 2014**

**STRAND: Earth and Space Sciences** **TOPIC: Our Universe** **YEAR: 10 (General + Extension)** **TERM: 4 (9 weeks)**

| TIME (WEEK) | STUDENT OUTCOMES | PRACTICALS | TEXT REFERENCE | ASSESSMENT/ HOMEWORK |
|---|---|---|---|---|
| 1 | **Introduction**<br><br>• Examine celestial objects, their relative size and scale, and how they fit into the cosmic scheme of things. | ScienceWeb - Lesson 1 (What's in our universe?) | | |
| 1-2 | **Origin of the Universe**<br><br>• Describe the competing theories on the origin of the universe - Creation, Steady State, Big Bang and Big Crunch.<br>• Identify the evidence that supports each theory.<br>• Understand that the Big Bang Theory is considered the best theory at the moment.<br>• Understand that our universe is considered to be expanding.<br>• Know the current distance to the edge of the "known" universe.<br>• Understand how the Doppler Effect gives an indication of the movement of galaxies. | Science Web - Lesson 3 The Big Bang Theory<br><br>Prac 2 p. 236 PS10 - Expanding universe.<br><br>Act 2.3 BI10 Workbook - Big Bang Model<br><br>Expt 2.1 p. 51 BI10 - Spectra<br><br>Invest. 6.5 p. 205 SQ10<br><br>Practivity 2.2 p. 52 BI10 - Doppler effect (use bull roarer) | SQ 10 Ch 6.3, 6.4<br><br>BI 10 Ch 7.1<br><br>PS 10 Ch 7.2<br><br>SA4 p. 3, 11 | Act 2.1 BI10 Workbook - History of the universe<br><br>Act 7.3 PS10 - History of cosmology<br><br>Research: NEO's + Origin of Universe |
| 3-4 | **The Universe**<br><br>• Describe the three different types of galaxies.<br>• Understand that constellations are patterns formed by stars.<br>• Name and locate common examples of constellations.<br>• Appreciate the vast distances involved in space travel.<br>• Understand the unit light year, and perform simple calculations using it.<br><br>***Extension***<br>• Describe how the parallax method is used to determine stellar distances. | ScienceWeb - Lesson 2 (Size + distance in space.)<br><br>Worksheets 10.1 + 10.3<br><br>TG Act 25: Southern Cross<br><br>Invest. 6.2 p. 199 SQ10<br><br>Prac 1 p. 73 SA4 | SQ 10 Ch 6.1<br><br>BI 10 Ch 7.2<br><br>PS 10 Ch 7.2<br><br>SA3 p. 4<br><br>SA4 p. 9-11 | Act 7.1 PS10 - Naming stars<br><br>Act 1.2 SA4 H/W book |

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| 5-6 | **Stars**<br><br>• Describe the layers and characteristics of our Sun.<br>• Understand the fusion process that produces energy in the Sun and relate it to Einstein's equation ($E = mc^2$).<br>• Understand that the colour of a star indicates its surface temperature.<br>• Understand the difference between apparent brightness and absolute brightness. | Invest. 6.3 p. 200 SQ10<br><br>Webquest: Sun Structure<br><br>Invest. 6.4 p. 203 SQ10 | SQ 10 Ch 6.2<br><br>BI 10 Ch 7.2<br><br>PS 10 Ch 7.2 | |
| 7-8 | **Life Cycles of Stars**<br><br>• Describe the life of a star from birth to death, dependent on its size.<br>• Understand the difference between a nova and supernova.<br>• Describe the characteristics of a white dwarf, neutron star and black hole.<br>• Describe the three possible pathways for the life of a star.<br><br>***Extension***<br>• Hertzsprung-Russell diagram for life cycles. | ScienceWeb - Lesson 4 (Life + death of stars.)<br><br>Worksheet 10.2<br><br>Webquest: Life Cycles<br><br>Practivity 2.5 p. 64 BI10 - Star profiles<br><br>Prac 3 p. 229 PS10 - Collapse of a neutron star | SQ 10 Ch 6.2<br><br>BI 10 Ch 7.2<br><br>PS 10 Ch 7.2<br><br>SA4 p. 3-7<br><br>SA3 p. 3-6<br><br>SA4 p. 19-22 | Act 7.2 PS10 - H-R Diagram<br><br>Act 1.1, 1.4 - 1.6 SA4 H/W book (Extension) |
| 9 | **Observing the Universe**<br><br>• Describe how radio telescopes and arrays of radio telescopes are used by astronomers and astrophysicists to observe distant parts of the universe.<br>• Explain how orbiting space telescopes are used to gather data from deep space and how they compare with Earth-based telescopes.<br>• Appreciate that the study of the universe and the exploration of space involves teams of specialists from different branches of science, engineering and technology. | ScienceWeb - Lesson 5 (Observing the universe.) | SQ 10 Ch 6.5 | Test : The Universe and Star Life Cycles |

**Texts** Science Quest 10 (SQ 10)
Big Ideas 10 (BI 10)
Pearson Science 10 (PS 10)
Science Aspects 3 + 4 (SA3, SA4)

**INTERNET SOURCES**

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| Great pictures of the universe | http://antwrp.gsfc.nasa.gov/apod/ap060506.html |
| Trinity College Astronomy | http://observatory.trinity.wa.edu.au/index.html |
| | (Very good links and photos.) |
| NASA | http://www.nasa.gov/home/ |
| | (Very good site for teachers and students.  Has terrific information on the Space Shuttle.) |